

Introduction to Bayesian Data Analysis

Lecture 6: Nonlinear Models: GLMs and the Shifted Log-normal

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Background

- GLMs are a big deal in frequentist stats
 - Do you use `glm` or `lme`?!?
- In Bayesian stats, we are used of thinking about our entire model more deliberately, including which model of the data to use.
- Yet, choosing between probability distributions for the data and interpreting the results requires a bit more thought than with linear models.

Overview

1. An Example

2. Model

- Generalized Linear Model
- Model of the Data
- Prior

3. Estimate the Model in `brms`

- Interpretation of the Results

4. Model fit

- Posterior prediction and Model comparison

5. The Shifted Log-normal

- Response times and the Shifted Log-normal Model
- Model in `brms`

An Example



Can mental health be predicted by daily habits and stress?

```
mentalhealth.dat <- read.csv("mental_health_dataset.csv")  
  
knitr::kable(head(mentalhealth.dat[, c(2,6,9,10,11,12)]))%>%  
  kable_styling(font_size = 16)
```

Age	Mental_Health_Condition	Stress_Level	Sleep_Hours	Work_Hours	Physical_Activity_Hours
36	No	Medium	7.1	46	5
34	Yes	Low	7.5	47	8
65	Yes	Low	8.4	58	10
34	No	Medium	9.8	30	2
22	Yes	Medium	4.9	62	5
64	Yes	High	6.3	34	0

An Example

"Do you have a mental illness?"



```
table(mentalhealth.dat$Mental_Health_Condition)
```

```
##
```

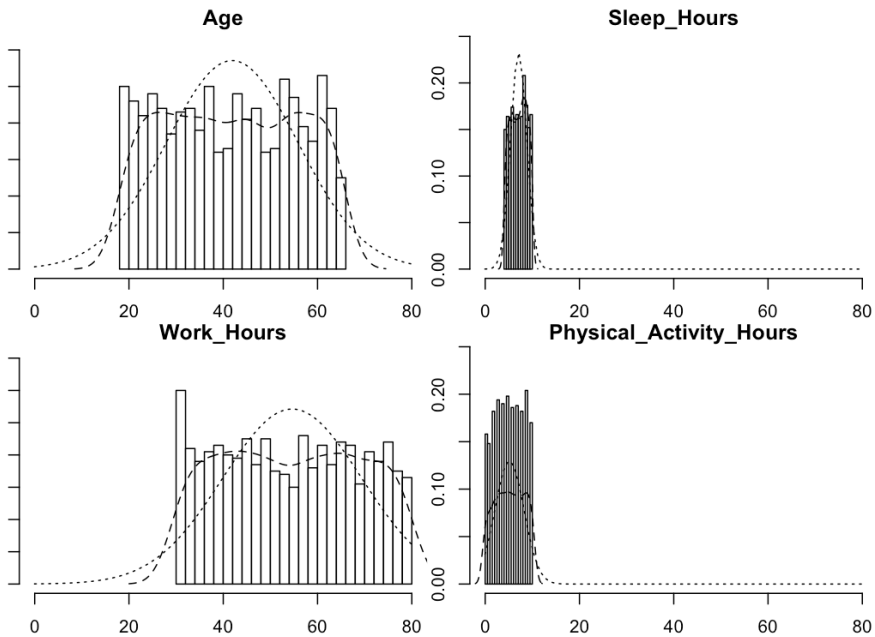
```
##   No  Yes
```

```
## 485 515
```

An Example



```
library("psych")  
multi.hist(mentalhealth.dat[, c(2, 10:12)])
```



The Model

What does a meaningful model for this data look like?

The Model

- Linear regression:

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2$$

- Extension:

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

Problems?

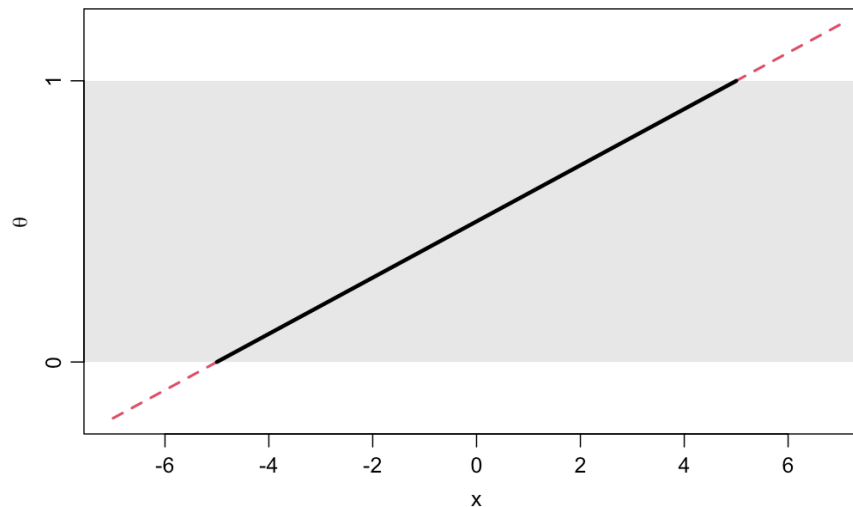
Y_i : Does the i th person have mental problems? $\rightarrow$ Yes/No
(1/0)

Statistical Model

Y_i : Does the i th person have mental problems? $\rightarrow$ Yes/No
(1/0)

- $Y_i \sim \text{Binomial}(n, \theta_i)$,
- $0 > \theta > 1, n = ?$

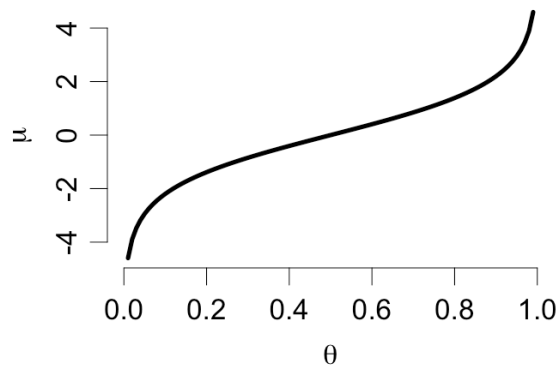
$$??? \theta_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n ???$$



Generalized Linear Model

Link Function $g(\cdot)$

- Goal: Connect the linear model with the parameter to be estimated (here: probability)
- For 0, 1 responses, the link function used is the logit transformation: $\mu_i = g(\theta_i) = \log \left(\frac{\theta_i}{1-\theta_i} \right)$



Generalized Linear Model

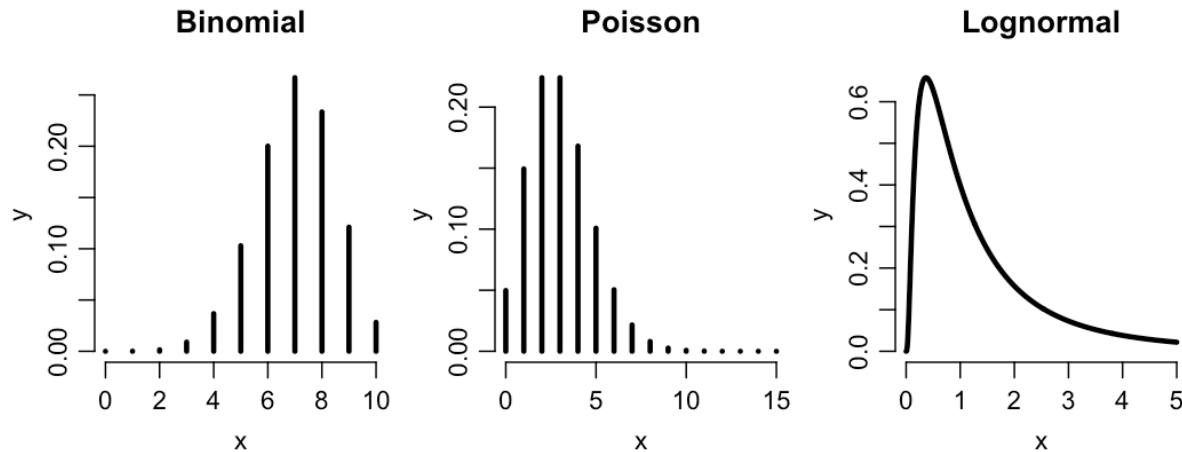
Using the link function, we go from the probability space into the "logit space" $\rightarrow \mu$ can be between $-\infty$ and ∞ .

Linear Model

$$\mu_i = \log \left(\frac{\theta_i}{1 - \theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

Generalized Linear Model

The generalized linear model specifies the probability distribution of a random variable and a link function that allows flexible use of linear regression models.

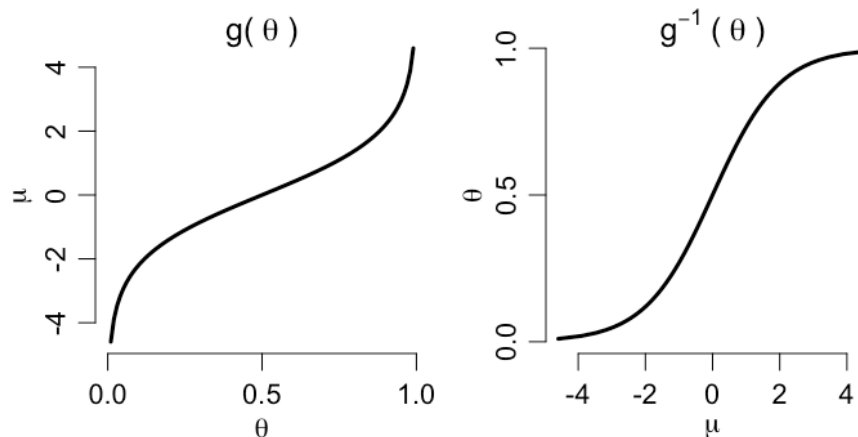


Generalized Linear Model

Inverse Function $g^{-1}(\cdot)$

- To get from μ_i back to θ_i we use the inverse of the logit function:

$$\theta_i = g^{-1}(\mu_i) = \frac{e^{\mu_i}}{1 + e^{\mu_i}}$$



Statistical Model

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- Predictors
 - $x_{i,1}$: Age
 - $x_{i,2}$: Stress level (high = 1, not high = 0)
 - $x_{i,3}$: Stress level (medium = 1, not medium = 0)
 - $x_{i,4}$: Sleep in hours
 - $x_{i,5}$: Work time per week in hours
 - $x_{i,6}$: Physical activity per week in hours
- $$\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$$

Statistical Model

How Does the Model Work?

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$
- Example:

$$\mu_i = -5 + x_{i,1}0.1 + x_{i,2}1 + x_{i,3}0.5 + x_{i,4}(-0.3) + \\ x_{i,5}0.1 + x_{i,6}(-0.2)$$

Statistical Model

$$\mu_i = -5 + x_{i,1}0.1 + x_{i,2}1 + x_{i,3}0.5 + x_{i,4}(-0.3) + \\ x_{i,5}0.1 + x_{i,6}(-0.2)$$

- Age = 40
- Stress level = medium
- Sleep in hours = 7
- Work hours per week = 32
- Physical activity per week = 8

$$\mu_i = -5 + 40 \times 0.1 + 0 \times 1 + 1 \times 0.5 + 7 \times (-0.3) + \\ 32 \times 0.1 + 8 \times (-0.2) = -1$$

Statistical Model

$$\begin{aligned}\mu_i = & -5 + 40 \times 0.1 + 0 \times 1 + 1 \times 0.5 + 7 \times (-0.3) + \\ & 32 \times 0.1 + 8 \times (-0.2) = -1\end{aligned}$$

The probability of mental problems is then

$$\theta_i = g^{-1}(-1) = \frac{e^{-1}}{1+e^{-1}} = 0.27$$

Statistical Model

- Age = 50
- Stress level = high
- Sleep in hours = 5
- Work hours per week = 60
- Physical activity per week = 0

$$\mu_i = -5 + 50 \times 0.1 + 1 \times 1 + 0 \times 0.5 + 5 \times (-0.3) + 60 \times 0.1 + 0 \times (-0.2) = 4.5$$

The probability of mental problems is then

$$\theta_i = g^{-1}(4.5) = \frac{e^{4.5}}{1+e^{4.5}} = 0.99$$

Prior

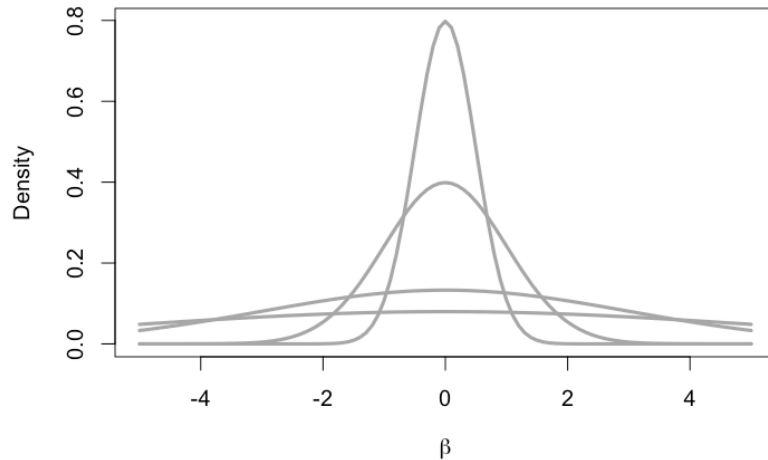
- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$

What parameters do we need in the model?

- All parameters!
 - $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$
 - Normal distribution
- What about σ ?

Prior

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$



- $x_{i,1}$: Age
- $x_{i,2}$: Stress level high
- $x_{i,3}$: Stress level med
- $x_{i,4}$: Sleep in hours
- $x_{i,5}$: Work hours
- $x_{i,6}$: Physical activity

Prior

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$

```
library(brms)
```

```
# default_prior(Mental_Health_Condition ~ Age +  
#               Stress_Level + Sleep_Hours + Work_Hours  
#               + Physical_Activity_Hours  
#               , family = bernoulli(link = "logit")  
#               , data = mentalhealth.dat)
```

```
bprior <- c(prior(normal(0, 3), class = Intercept)  
            , prior(normal(0, 0.5), class = b))
```

Estimating the Model

Estimating the Model

```
model.1 <- brm(Mental_Health_Condition ~ Age +
               Stress_Level + Sleep_Hours + Work_Hours
               + Physical_Activity_Hours
               , family = bernoulli(link = "logit")
               , data = mentalhealth.dat
               , prior = bprior)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
##
```

```
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
```

```
## Chain 1:
```

```
## Chain 1: Gradient evaluation took 2.1e-05 seconds
```

```
## Chain 1: 1000 transitions using 10 leapfrog steps per transition
would take 0.21 seconds.
```

```
## Chain 1: Adjust your expectations accordingly!
```

```
## Chain 1:
```

```
## Chain 1:
```

```
## Chain 1: Iteration:      1 / 2000 [  0%]   (Warmup)
```

```
## Chain 1: Iteration:    200 / 2000 [ 10%]   (Warmup)
```

```
## Chain 1: Iteration:   1000 / 2000 [ 50%]   (Warmup)
```

Estimating the Model

```
summary(model.1)
```

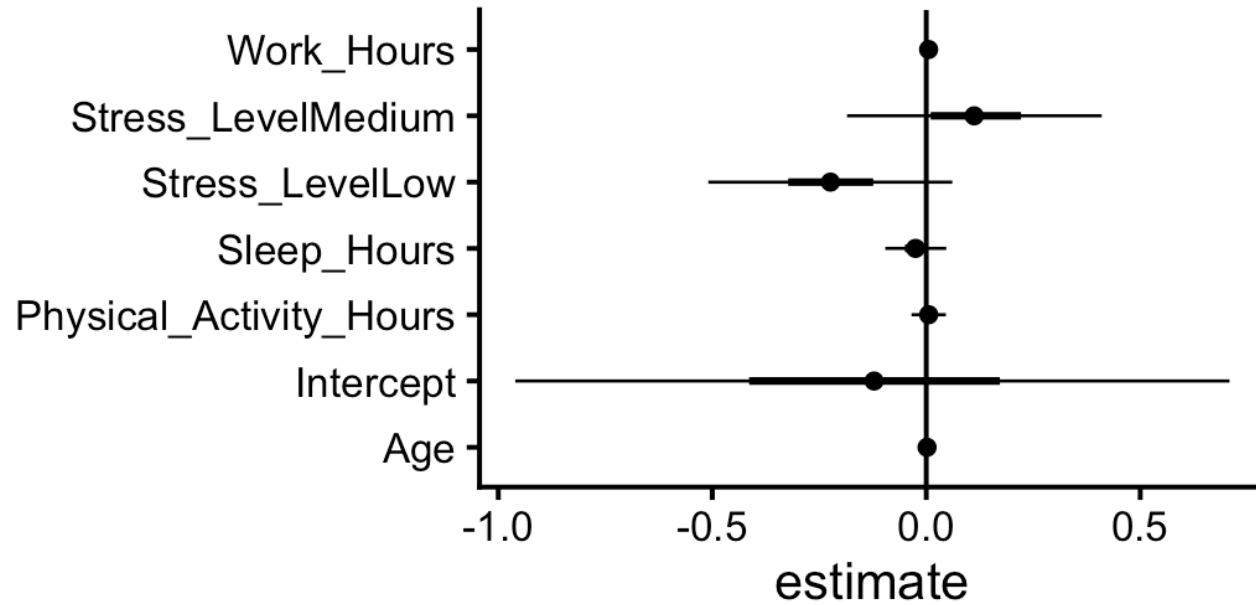
```
## Family: bernoulli
## Links: mu = logit
## Formula: Mental_Health_Condition ~ Age + Stress_Level + Sleep_Hours +
Work_Hours + Physical_Activity_Hours
## Data: mentalhealth.dat (Number of observations: 1000)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Bulk_ESS					
## Intercept	-0.12	0.44	-0.96	0.71	1.00
5493					
## Age	0.00	0.00	-0.01	0.01	1.00
6869					
## Stress_LevelLow	-0.22	0.14	-0.51	0.06	1.00
4119					
## Stress_LevelMedium	0.11	0.15	-0.19	0.41	1.00
4493					
## Sleep_Hours	-0.03	0.04	-0.10	0.05	1.00
5346					

Estimating the Model

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.12	0.44	-0.96	0.71	1	5492.86	3268.71
Age	0.00	0.00	-0.01	0.01	1	6869.45	3209.92
Stress_LevelLow	-0.22	0.14	-0.51	0.06	1	4118.97	3055.14
Stress_LevelMedium	0.11	0.15	-0.19	0.41	1	4492.62	2906.46
Sleep_Hours	-0.03	0.04	-0.10	0.05	1	5345.52	3105.04
Work_Hours	0.01	0.00	0.00	0.01	1	6541.06	3121.18
Physical_Activity_Hours	0.01	0.02	-0.03	0.05	1	5873.64	3013.60

Estimating the Model

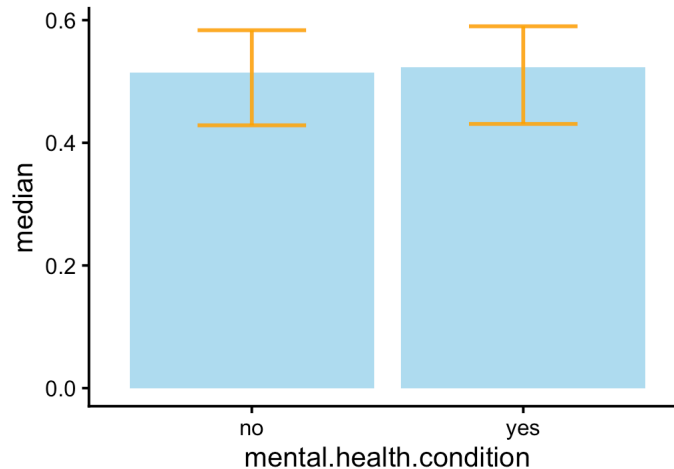


Posterior Prediction

Does the model accurately predict the response category for mental disorder?

```
post.pred <- posterior_predict(model.1) #as before
post.pred.mean <- colMeans(post.pred) #as before

# Median and prediction interval for y
post.pred.dat <- tapply(post.pred.mean
                        , mentalhealth.dat$Mental_Health_Condition
                        , quantile, probs = c(.025, .5, .975))
```



Posterior Prediction

```
# tapply gives a matrix, for ggplot we need a data frame
post.pred.dat <- as.data.frame(rbind(post.pred.dat$Yes
                                     , post.pred.dat$No))

# Naming the columns and rows
colnames(post.pred.dat) <- c("lower", "median", "upper")
post.pred.dat$mental.health.condition <- c("yes", "no")

ggplot(post.pred.dat) +
  geom_bar(aes(x = mental.health.condition
               , y = median)
           , stat = "identity"
           , fill = "skyblue"
           , alpha = 0.7) +
  geom_errorbar(aes(x = mental.health.condition
                    , ymin=lower
                    , ymax=upper)
                , width=0.4, colour="orange"
                , alpha=0.9, linewidth=1.3)+
  theme_classic(base_size = 20)
```

Model Comparison

Is the model informative overall?

→ Comparison with a model without predictors

```
model.1 <- brm(Mental_Health_Condition ~ Age +  
               Stress_Level + Sleep_Hours + Work_Hours  
               + Physical_Activity_Hours  
               , family = bernoulli(link = "logit")  
               , data = mentalhealth.dat  
               , prior = bprior  
               , save_pars = save_pars(all = TRUE)  
               , iter = 11000  
               , warmup = 1000  
               , silent = 2  
               , refresh = 0)
```

```
bprior <- c(prior(normal(0, 3), class = Intercept))
```

```
model.0 <- brm(Mental_Health_Condition ~ 1  
               , family = bernoulli(link = "logit")  
               , data = mentalhealth.dat  
               , prior = bprior
```

Model Comparison

```
bayes_factor(model.0, model.1)
```

```
## Warning: effective sample size cannot be calculated, has been  
replaced by
```

```
## number of samples.
```

```
## Iteration: 1
```

```
## Iteration: 2
```

```
## Iteration: 3
```

```
## Iteration: 4
```

```
## Iteration: 1
```

```
## Iteration: 2
```

```
## Iteration: 3
```

```
## Iteration: 4
```

```
## Estimated Bayes factor in favor of model.0 over model.1:  
1456695.93664
```

Model Comparison

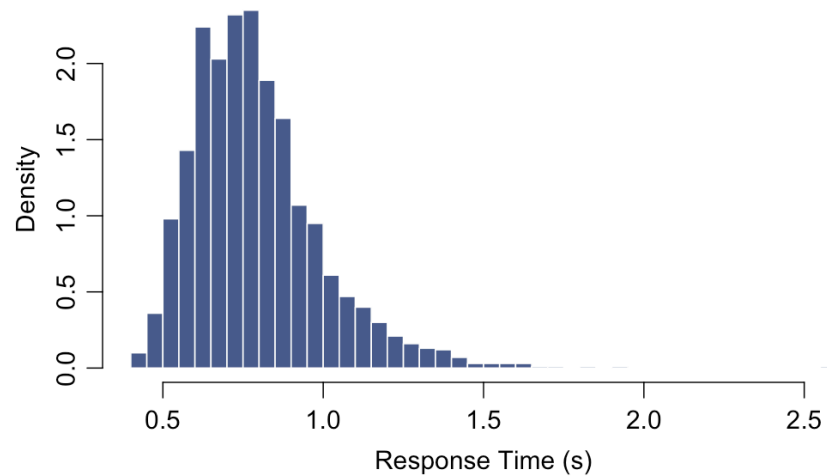
- Strong evidence against effects of all predictors.
- Prevalence for a mental disorder cannot be predicted by age, life habits, and stress (for this dataset).
- Possible reasons?



The shifted log-normal model

Response times

Reaction times (RTs) are one of the most common outcome variables in psychology



Response times

What's so special?

- RTs are **always positive**
- The distribution is **right-skewed**: most responses cluster close together, but there is a long tail of slow responses that stretches to the right
- There appears to be a **minimum time** below which responses almost never occur

Response times

Why?

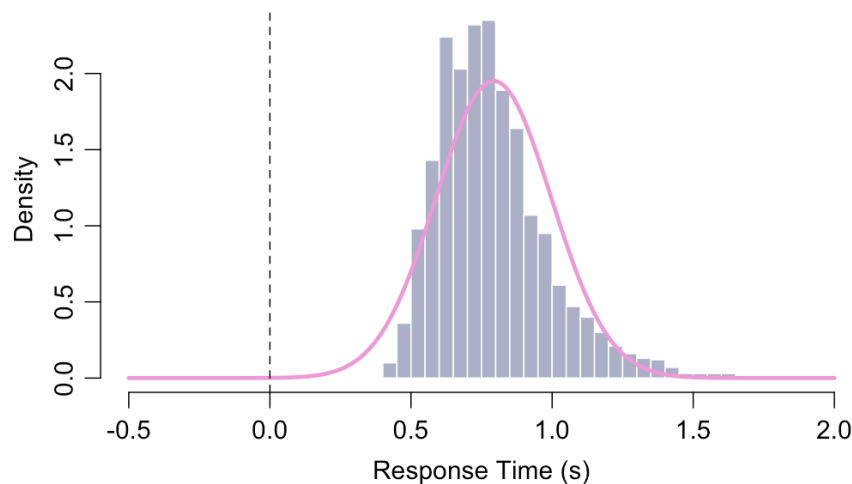
- A response can't happen before the stimulus has been perceived, processed, and a motor response prepared and executed, which means there is a **hard lower bound** (often called t_0 , encoding & motor time)
- Most trials are quick and similarly responded to but sometimes our attention lapses or motor execution is delayed (e.g., because we are tired), producing a **long right tail**

Response times

- Because of these features a **symmetric, unbounded** distribution (like the Normal) is a poor fit

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

would assign probability to $Y_i < 0$, and cannot reproduce the skew



The Shifted Log-normal

We need a distribution for the data that:

1. Puts mass only for values above this encoding time (τ or non-decision time)
2. And is right-skewed

$Y_i - \tau$ could be log-normally distributed. Then Y_i is bigger than τ . And since the log-normal is right-skewed, Y_i is right-skewed then as well.

This is the **shifted log-normal** model:

$$Y_i - \tau \sim \text{Log-Normal}(\mu_i, \sigma^2)$$

The Shifted Log-normal

Link Function

- Because we already are on the $\log(Y_i - \tau)$ scale, the link function is:

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

- μ_i is the (linear predictor for the) mean of the *log* RT above the shift and not the mean RT

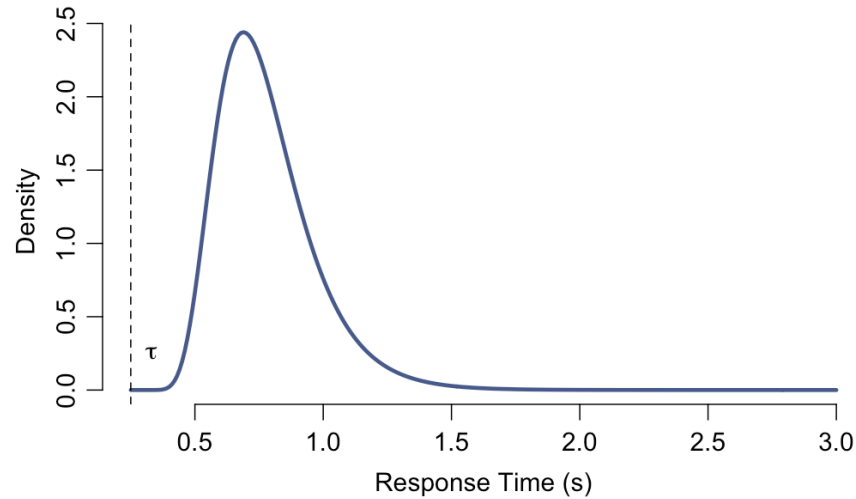
The Shifted Log-normal

- To get back to the RT scale, we need the to inverse transform:

$$\mathbb{E}[Y_i] = \tau + \exp\left(\mu_i + \frac{\sigma^2}{2}\right)$$

- τ is also a parameter that needs to be estimated, it has the constraint $\tau < \min(Y_i)$, since nobody can respond before the shift

The Shifted Log-normal



The shift τ moves the whole distribution so that it starts after encoding time.

The Shifted Log-normal

Statistical Model

- $Y_i - \tau \sim \text{Log-Normal}(\mu_i, \sigma^2)$
- $\mu_i = \beta_0 + x_{i,1}\beta_1 + \dots + x_{i,n}\beta_n$
- Parameters to estimate:
 - $\beta_0, \dots, \beta_n$ (linear predictor)
 - σ (residual SD on log scale)
 - τ (shift / non-decision time)

The Shifted Log-normal

- Priors still need to respect the parameter space:
 - Normal priors on the β s (it's a log-RT scale, so this should be small, e.g. $\beta \sim \text{Normal}(0, 0.5)$)
 - A positive prior on σ (e.g. half-Normal or Exponential)
 - A prior on τ bounded to below $\min(Y_i)$

The Shifted Log-normal

```
library(brms)

bprior <- c(prior(normal(0, 3), class = Intercept)
            , prior(normal(0, 0.5), class = b)
            , prior(normal(0, 1), class = sigma)
            , prior(uniform(0, min_RT), class = ndt, ub = min_RT))

model.rt <- brm(RT ~ Age + Stress_Level + Sleep_Hours + Work_Hours
                + Physical_Activity_Hours
                , family = shifted_lognormal()
                , data = rt.dat
                , prior = bprior)
```

`brms` calls the shift parameter `ndt` (non-decision time), that's τ .

The Shifted Log-normal

Interpretation

- Coefficients β_j are on the **log-RT** scale: one unit increase in x_j multiplies the (shifted) RT by $\exp(\beta_j)$
- $\exp(\beta_j) > 1$: predictor is associated with **slower** responses
- $\exp(\beta_j) < 1$: predictor is associated with **faster** responses
- τ (**ndt**) is interpretable in seconds, it's the estimated bound below which no response is possible

